

**BACHELOR OF COMPUTER
APPLICATIONS (BCA) (REVISED)**

Term-End Examination

June, 2023

**BCS-041 : FUNDAMENTALS OF COMPUTER
NETWORKS**

Time : 3 Hours

Maximum Marks : 100

Note : *Question number 1 is compulsory. Attempt any **three** questions from the rest. Use of calculator is allowed.*

1. (a) Differentiate between private key and public key cryptography. Give suitable example of each. 6
- (b) Briefly discuss the concept of frequency shift keying and phase shift keying. Give an application of each. 6

- (c) What is count to infinity problem in distance vector routing protocol ? How does it happen ? Explain with an example. 10
- (d) Briefly discuss the functions of Layer-2 switch and Layer-3 switch. Compare Switch with Hub. 6
- (e) How does pure ALOHA differ from slotted ALOHA ? Explain. 6
- (f) What are virtual circuits ? Discuss the effect of router failure in virtual circuits. 6
2. (a) Differentiate between ARP and RARP. Explain the working of ARP using a diagram. 10
- (b) How does classful addressing differ from classless addressing ? How does classless addressing result in decrease in the table size ? 10
3. (a) Explain the concept *Go-Back-N sliding window protocol* with a suitable example and diagram. 10

- (b) What is Fragmentation ? Explain why IPv4 and IPv6 protocols need to fragment some packets. 10
4. (a) Write the step-by-step working of link state routing. Also, compare it with distance vector routing. 10
- (b) Discuss the concept of sliding window protocol with the help of an example. Also, explain how piggybacking technique works. 10
5. (a) What is IGMP ? Draw the header fields of IGMP. Also, explain the significance of each field. 10
- (b) Write the significance and usage of the following networking devices : 2×5=10
- (i) Repeater
 - (ii) Bridges
 - (iii) Switches
 - (iv) Gateways
 - (v) Networks interface card